

Trigonometry Problems

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1. Prove that:

$$\cos \frac{2\pi}{13} + \cos \frac{6\pi}{13} + \cos \frac{8\pi}{13} = \frac{\sqrt{13} - 1}{4}$$

2. Prove that $2 \left(\cos \frac{4\pi}{19} + \cos \frac{6\pi}{19} + \cos \frac{10\pi}{19} \right)$ is a root of the equation:

$$\sqrt{4 + \sqrt{4 + \sqrt{4 - x}}} = x$$

3. Prove that

$$\sqrt{\frac{1}{2} + \frac{1}{2} \sqrt{\frac{1}{2} + \frac{1}{2} \sqrt{\frac{1}{2} + \frac{1}{2} \cos 8\theta}}} = \cos \theta$$

4. Prove that

$$\sin^4 \left(\frac{\pi}{8} \right) + \sin^4 \left(\frac{3\pi}{8} \right) + \sin^4 \left(\frac{5\pi}{8} \right) + \sin^4 \left(\frac{7\pi}{8} \right) = \frac{3}{2}$$

5. Prove that

$$\cos x \cdot \cos \left(\frac{x}{2} \right) \cdot \cos \left(\frac{x}{4} \right) \cdot \cos \left(\frac{x}{8} \right) = \frac{\sin 2x}{16 \sin \left(\frac{x}{8} \right)}$$

6. Prove that

$$64 \cdot \sin 10^\circ \cdot \sin 20^\circ \cdot \sin 30^\circ \cdot \sin 40^\circ \cdot \sin 50^\circ \cdot \sin 60^\circ \cdot \sin 70^\circ \cdot \sin 80^\circ \cdot \sin 90^\circ = \frac{3}{4}$$

7. Find x if

$$\sin x = \tan 12^\circ \cdot \tan 48^\circ \cdot \tan 54^\circ \cdot \tan 72^\circ.$$

8. Solve the following equations in $\mathbb{R}$:

- $\sin 9x + \sin 5x + 2 \sin^2 x = 1$
- $\cos 5x \cdot \cos 3x - \sin 3x \cdot \sin x = \cos 2x$
- $\cos 5x + \cos 3x + \sin 5x + \sin 3x = 2 \cdot \cos\left(\frac{\pi}{4} - 4x\right)$
- $\sin x + \cos x - \sin x \cdot \cos x = -1$
- $\sin 2x - \sqrt{3} \cos 2x = 2$

9. Prove following equations:

- $\sin\left(\frac{2\pi}{7}\right) + \sin\left(\frac{4\pi}{7}\right) - \sin\left(\frac{6\pi}{7}\right) = 4 \sin\left(\frac{\pi}{7}\right) \cdot \sin\left(\frac{3\pi}{7}\right) \cdot \sin\left(\frac{5\pi}{7}\right)$
- $\cos\left(\frac{\pi}{13}\right) + \cos\left(\frac{3\pi}{13}\right) + \cos\left(\frac{5\pi}{13}\right) + \cos\left(\frac{7\pi}{13}\right) + \cos\left(\frac{9\pi}{13}\right) + \cos\left(\frac{11\pi}{13}\right) = \frac{1}{2}$
- $\forall k \in \mathbb{N} : \cos\left(\frac{\pi}{2k+1}\right) + \cos\left(\frac{3\pi}{2k+1}\right) + \cdots + \cos\left(\frac{(2k-1)\pi}{2k+1}\right) = \frac{1}{2}$
- $\sin\left(\frac{\pi}{7}\right) + \sin\left(\frac{2\pi}{7}\right) + \sin\left(\frac{3\pi}{7}\right) = \frac{1}{4} \cdot \cot\left(\frac{\pi}{4}\right)$

10. Show that

$$\cos \frac{\pi}{n} + \cos \frac{2\pi}{n} + \cdots + \cos \frac{n\pi}{n} = -1.$$

11. Show that $\cos a + \cos 3a + \cos 5a + \cdots + \cos(2n-1)a = \frac{\sin 2na}{2 \sin a}$.

12. Show that $\sin a + \sin 3a + \sin 5a + \cdots + \sin(2n-1)a = \frac{\sin^2 na}{\sin a}$.

13. Calculate

$$(\tan 1^\circ)^2 + (\tan 2^\circ)^2 + (\tan 3^\circ)^2 + \cdots + (\tan 89^\circ)^2.$$

14. Prove that $\cot^2 \frac{\pi}{7} + \cot^2 \frac{2\pi}{7} + \cot^2 \frac{3\pi}{7} = 5$.

15. Show that $\tan \frac{\pi}{7} \tan \frac{2\pi}{7} \tan \frac{3\pi}{7} = \sqrt{7}$.

16. $\cos\left(\frac{2\pi}{7}\right)$, $\cos\left(\frac{4\pi}{7}\right)$ and $\cos\left(\frac{6\pi}{7}\right)$ are the roots of an equation of the form $ax^3 + bx^2 + cx + d = 0$ where a, b, c, d are integers. Determine a, b, c and d .

***17.** Find the value of the sum

$$\sqrt[3]{\cos \frac{2\pi}{7}} + \sqrt[3]{\cos \frac{4\pi}{7}} + \sqrt[3]{\cos \frac{6\pi}{7}}.$$

18. Solve the equation

$$2 \sin^4 x (\sin 2x - 3) - 2 \sin^2 x (\sin 2x - 3) - 1 = 0.$$

19. Express the sum of the following series in terms of $\sin x$ and $\cos x$.

$$\sum_{k=0}^n (2k+1) \sin^2 \left(x + \frac{k}{2} \pi \right)$$

20. Find the smallest positive integer N for which

$$\frac{1}{\sin 45^\circ \cdot \sin 46^\circ} + \frac{1}{\sin 47^\circ \cdot \sin 48^\circ} + \cdots + \frac{1}{\sin 133^\circ \cdot \sin 134^\circ} = \frac{1}{\sin N^\circ}.$$

21. Find the value of

$$\frac{\sin 40^\circ + \sin 80^\circ}{\sin 110^\circ}.$$

22. Evaluate the sum

$$S = \tan 1^\circ \cdot \tan 2^\circ + \tan 2^\circ \cdot \tan 3^\circ + \tan 3^\circ \cdot \tan 4^\circ + \cdots + \tan 2004^\circ \cdot \tan 2005^\circ.$$

23. Solve the equation :

$$\sqrt{3} \sin x (\cos x - \sin x) + (2 - \sqrt{6}) \cos x + 2 \sin x + \sqrt{3} - 2\sqrt{2} = 0.$$

24. Let $f(x) = \frac{1}{\sin \frac{\pi x}{7}}$. Prove that $f(3) + f(2) = f(1)$.

25. Suppose that real numbers x, y, z satisfy

$$\frac{\cos x + \cos y + \cos z}{\cos (x + y + z)} = \frac{\sin x + \sin y + \sin z}{\sin (x + y + z)} = p$$

Prove that

$$\cos(x+y) + \cos(y+z) + \cos(x+z) = p.$$

26. Solve for $\theta, 0 \leq \theta \leq \frac{\pi}{2}$:

$$\sin^5 \theta + \cos^5 \theta = 1.$$

27. For $x, y \in [0, \frac{\pi}{3}]$ prove that $\cos x + \cos y \leq 1 + \cos xy$.

28. Prove that among any four distinct numbers from the interval $(0, \frac{\pi}{2})$ there are two, say x, y , such that:

$$8 \cos x \cos y \cos(x-y) + 1 > 4(\cos^2 x + \cos^2 y).$$

29. Let $B = \frac{\pi}{7}$. Prove that

$$\tan B \cdot \tan 2B + \tan 2B \cdot \tan 4B + \tan 4B \cdot \tan B = -7.$$

30. a) Calculate

$$\frac{1}{\cos \frac{6\pi}{13}} - 4 \cos \frac{4\pi}{13} - 4 \cos \frac{5\pi}{13} = ?$$

b) Prove that

$$\tan \frac{\pi}{13} + 4 \sin \frac{4\pi}{13} = \tan \frac{3\pi}{13} + 4 \sin \frac{3\pi}{13}$$

c) Prove that

$$\tan \frac{2\pi}{13} + 4 \sin \frac{6\pi}{13} = \tan \frac{5\pi}{13} + 4 \sin \frac{2\pi}{13}$$

31. Prove that if α, β are angles of a triangle and $(\cos^2 \alpha + \cos^2 \beta)(1 + \tan \alpha \cdot \tan \beta) = 2$, then $\alpha + \beta = 90^\circ$.

32. Let $a, b, c, d \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ be real numbers such that $\sin a + \sin b + \sin c + \sin d = 1$ and $\cos 2a + \cos 2b + \cos 2c + \cos 2d \geq \frac{10}{3}$. Prove that $a, b, c, d \in [0, \frac{\pi}{6}]$

33. Find all integers m, n for which we have $\sin^m x + \cos^n x = 1$, for all x .

34. Prove that $\tan 55^\circ \cdot \tan 65^\circ \cdot \tan 75^\circ = \tan 85^\circ$.

35. Prove that $\frac{4 \cos 12^\circ + 4 \cos 36^\circ + 1}{\sqrt{3}} = \tan 78^\circ$.

36. Prove that

$$\sqrt{4 + \sqrt{4 + \sqrt{4 - \sqrt{4 + \sqrt{4 + \sqrt{4 - \dots}}}}} = 2 \left(\cos \frac{4\pi}{19} + \cos \frac{6\pi}{19} + \cos \frac{10\pi}{19} \right).$$

The signs: $++-++-++-++-\dots$

37. For reals x, y Prove that $\cos x + \cos y + \sin x \sin y \leq 2$.

38. Solve the equation in real numbers

$$\sqrt{7 + 2\sqrt{7 - 2\sqrt{7 - 2x}}} = x.$$

39. Let A, B, C be three angles of triangle ABC . Prove that

$$(1 - \cos A)(1 - \cos B)(1 - \cos C) \geq \cos A \cos B \cos C.$$

40. Solve the equation

$$\sin^3(x) - \cos^3(x) = \sin^2(x).$$

41. Find $S_n = \sum_{k=1}^n \sin^2 k\theta$ for $n \geq 1$

42. Prove the following without using induction:

$$\cos x + \cos 2x + \dots + \cos nx = \frac{\cos \frac{n+1}{2}x \cdot \sin \frac{n}{2}x}{\sin \frac{x}{2}}.$$

43. Evaluate:

$$\sin \theta + \frac{1}{2} \cdot \sin 2\theta + \frac{1}{2^2} \cdot \sin 3\theta + \frac{1}{2^3} \cdot \sin 4\theta + \dots$$

44. Compute

$$\sum_{k=1}^{n-1} \csc^2 \left(\frac{k\pi}{n} \right).$$

45. Prove that

$$\bullet \quad \tan \theta + \tan \left(\theta + \frac{\pi}{n} \right) + \tan \left(\theta + \frac{2\pi}{n} \right) + \dots + \tan \left[\theta + \frac{(n-1)\pi}{n} \right] = -n \cot \left(n\theta + \frac{n\pi}{2} \right).$$

- $\cot \theta + \cot \left(\theta + \frac{\pi}{n} \right) + \cot \left(\theta + \frac{2\pi}{n} \right) + \cdots + \cot \left[\theta + \frac{(n-1)\pi}{n} \right] = n \cot n\theta.$

46. Calculate

$$\sum_{n=1}^{\infty} 2^{2n} \sin^4 \frac{a}{2^n}.$$

47. Compute the following sum:

$$\tan 1^\circ + \tan 5^\circ + \tan 9^\circ + \cdots + \tan 177^\circ.$$

48. Show that for any positive integer $n > 1$,

- $\sum_{k=0}^{n-1} \cos \frac{2\pi k^2}{n} = \frac{\sqrt{n}}{2} \left(1 + \cos \frac{n\pi}{2} + \sin \frac{n\pi}{2} \right)$
- $\sum_{k=0}^{n-1} \sin \frac{2\pi k^2}{n} = \frac{\sqrt{n}}{2} \left(1 + \cos \frac{n\pi}{2} - \sin \frac{n\pi}{2} \right)$

49. Evaluate the product

$$\prod_{k=1}^n \tan \frac{k\pi}{2(n+1)}.$$

50. Prove that, $\sum_{k=1}^n (-1)^{k-1} \cot \frac{(2k-1)\pi}{4n} = n$ for even n .

51. Prove that $\sum_{k=1}^n \cot^2 \left\{ \frac{(2k-1)\pi}{2n} \right\} = n(2n-1).$

52. Prove that $\sum_{k=1}^n \cot^4 \left(\frac{k\pi}{2n+1} \right) = \frac{n(2n-1)(4n^2+10n-9)}{45}.$

53. Let x be a real number with $0 < x < \pi$. Prove that, for all natural numbers n , the sum $\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \cdots + \frac{\sin(2n-1)x}{2n-1}$ is positive.